Ride Sharing Analytics & Driver Performance pipeline

A

Project Report

Submitted By

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1.Project Overview

The Ride Sharing Data Engineering Pipeline is a PySpark-based project that processes ride-sharing data using the Medallion Architecture (Bronze, Silver, and Gold). It uses three datasets: Drivers, Trips, and Trip Logs. The Bronze layer ingests raw CSV data and stores it in Parquet format. The Silver layer cleans the data, removes duplicates, joins the datasets, and creates new columns such as Trip Duration, Completion Status, and Driver Performance. The Gold layer generates business insights, including Total Revenue, Top Drivers, Revenue by City, Cancellation Analysis, and Popular Pickup Locations. This project demonstrates how raw data can be transformed into clean and meaningful information for business analysis using PySpark.

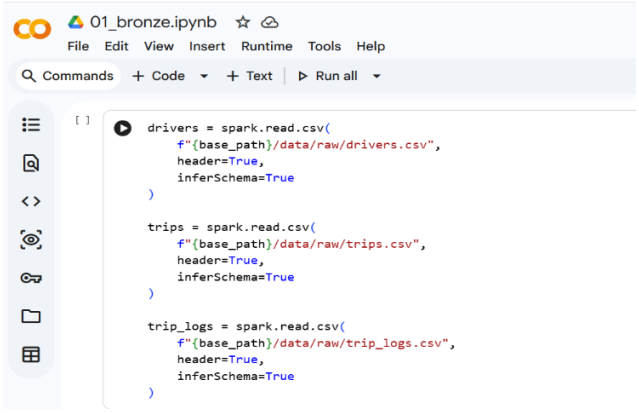
2. Technologies Used

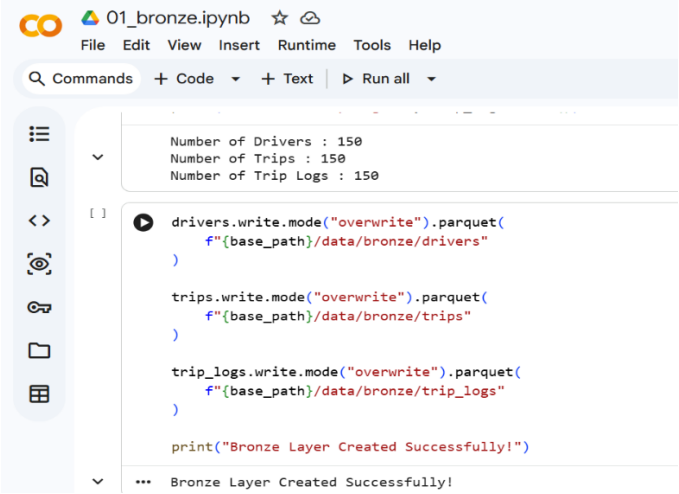
- Python
- PySpark
- Google Colab
- Google Drive
- Git & GitHub

3. Project Workflow

i.Bronze Layer

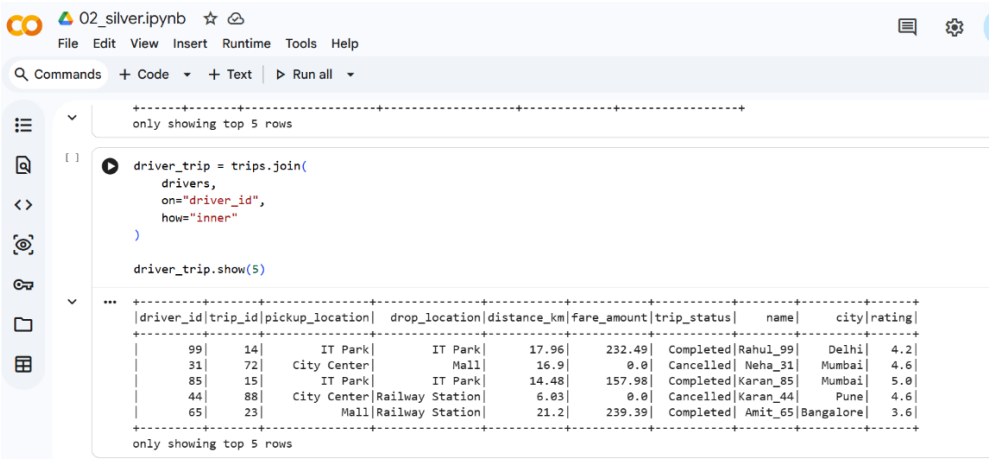
- Reads raw CSV files using PySpark.
- Converts raw data into Parquet format.
- Stores data in the Bronze layer.

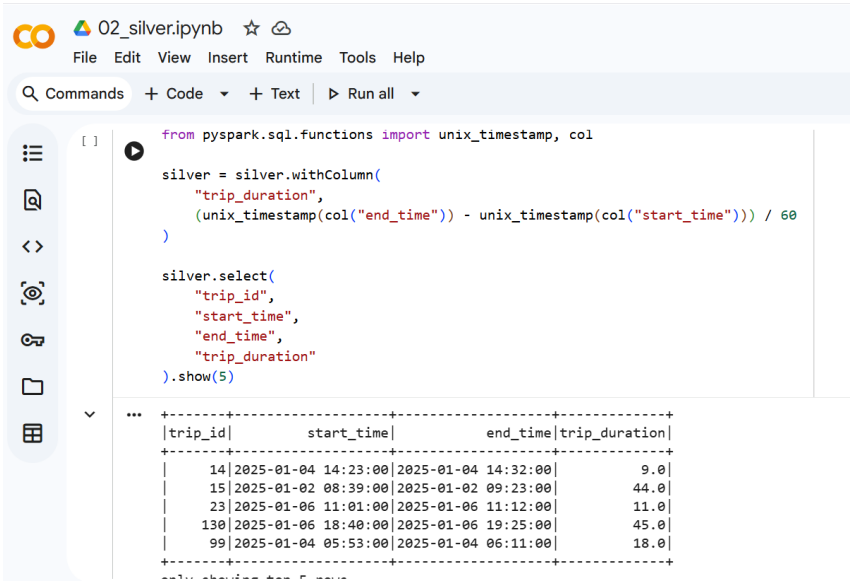


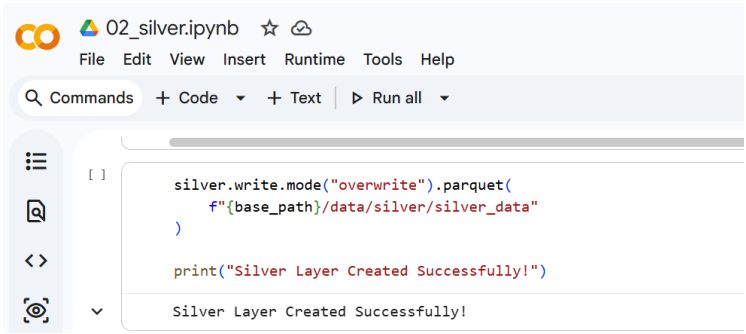


ii.Silver Layer

- Reads Bronze data.
- Removes duplicates and handles missing values.
- Joins the datasets.
- Calculates Trip Duration.
- Creates Completion Status and Driver Performance columns.
- Stores cleaned data in the Silver layer.

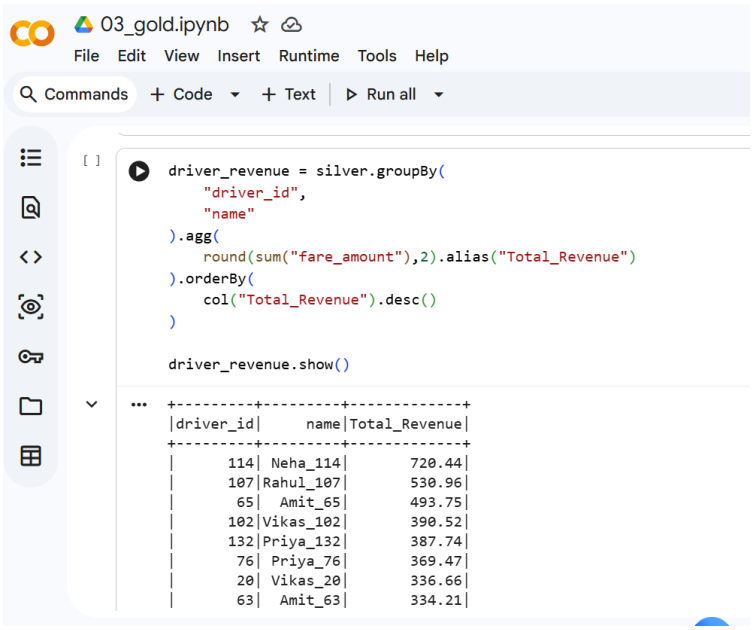


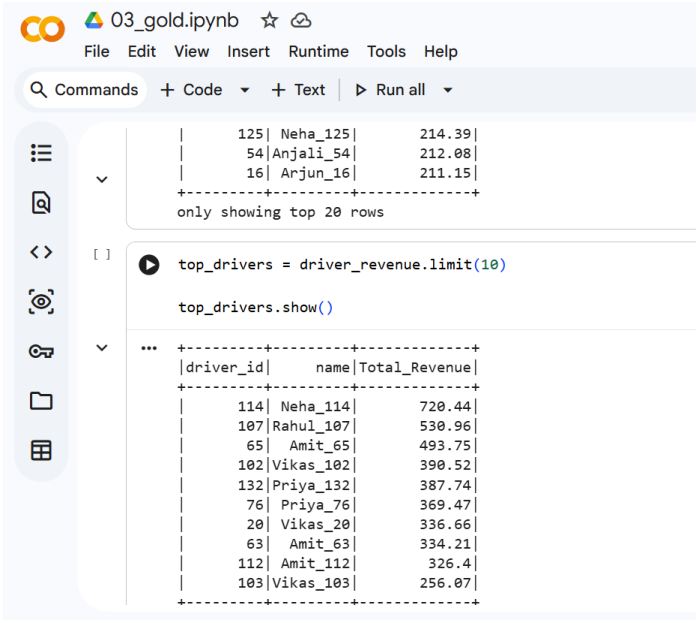




iii.Gold Layer

- Reads Silver data.
- Performs business analysis.
- Calculates Total Revenue, Top Drivers, Revenue by City, Cancellation Analysis, and Popular Pickup Locations.
- Stores the final business-ready data in the Gold layer.


03_gold.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
[ ]
driver_revenue = silver.groupBy(
"driver_id",
"name"
).agg(
round(sum("fare_amount"),2).alias("Total_Revenue")
).orderBy(
col("Total_Revenue").desc()
)
driver_revenue.show()
+---------+---------+-------------+
|driver_id| name|Total_Revenue|
+---------+---------+-------------+
114	Neha_114	720.44
107	Rahul_107	530.96
65	Amit_65	493.75
102	Vikas_102	390.52
132	Priya_132	387.74
76	Priya_76	369.47
20	Vikas_20	336.66
63	Amit_63	334.21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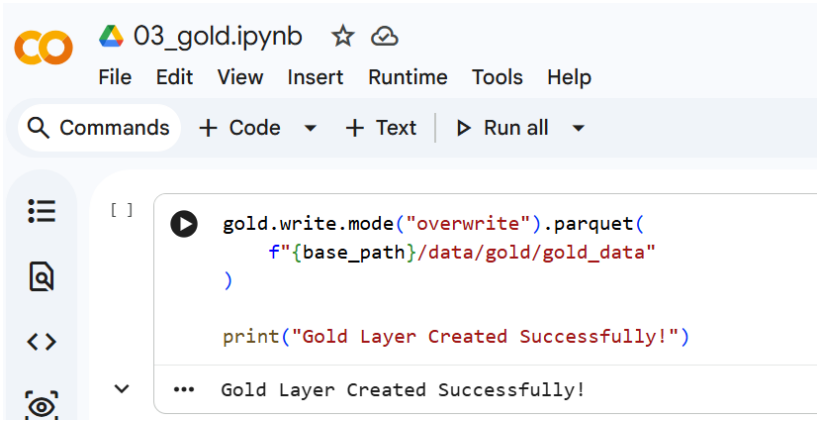
125	Neha_125	214.39
54	Anjali_54	212.08
16	Arjun_16	211.15
+---------+---------+-------------+
only showing top 20 rows
[ ]
top_drivers = driver_revenue.limit(10)
top_drivers.show()
+---------+---------+-------------+
|driver_id| name|Total_Revenue|
+---------+---------+-------------+
114	Neha_114	720.44
107	Rahul_107	530.96
65	Amit_65	493.75
102	Vikas_102	390.52
132	Priya_132	387.74
76	Priya_76	369.47
20	Vikas_20	336.66
63	Amit_63	334.21
112	Amit_112	326.4
103	Vikas_103	256.07
+---------+---------+-------------+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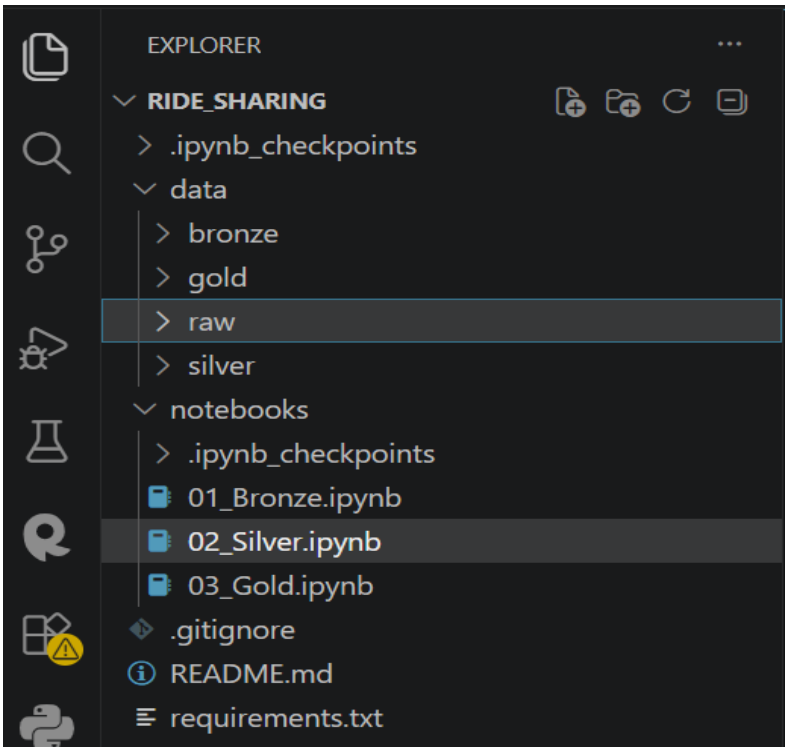
[ ]
gold.write.mode("overwrite").parquet(
f"{base_path}/data/gold/gold_data"
)
print("Gold Layer Created Successfully!")
Gold Layer Created Successfully!

4.Project Structure



5. Conclusion

This project successfully implements the Medallion Architecture using PySpark. The raw data is transformed into clean and structured datasets, and the Gold layer provides meaningful business insights that can support decision-making in a ride-sharing system.